LOGISTICS OPERATIONS ANALYTICS

Project Report — Final Submission

Business Analytics Team | Third-Party Logistics (3PL) Company | Nationwide Freight Operations

Project Title	Logistics Operations Analytics
Department	Business Analytics
Prepared By	Ashirbad Routray
Submitted To	Project Manager
Report Date	March 10, 2026
Project Deadline	March 10, 2026
Status	COMPLETED
Classification	CONFIDENTIAL — INTERNAL USE ONLY

CONFIDENTIAL — FOR INTERNAL USE ONLY

Table of Contents

	Title	Page No.
1.	Executive Summary……..	03
2.	Project Overview…….	04
4.	Data Architecture & Sources……	05
5.	Analytical Framework…….	06
6.	Module 1 — Driver Performance Insights & Findings……	07
7.	Module 2 — Route Profitability Assessment & Findings……..	09
8.	Module 3 — Fleet Utilization Monitoring & Findings…….	12
9.	Module 4 — Maintenance Cost Impact Analysis & Findings…….	14
10.	Module 5 — Fuel Efficiency Analysis & Findings……..	16
11.	Module 6 — Customer Revenue Analysis & Findings……..	18
12.	Module 7 — Safety Risk Assessment & Findings…….	20
13.	Module 8 — Seasonal Demand Pattern Analysis & Findings…….	23
14.	Cross-Module Key Findings……..	26
15.	Strategic Recommendations…….	28
16.	Conclusion…….	30

1. Executive Summary

This report constitutes the final deliverable of the Logistics Operations Analytics project commissioned by the Project Manager and executed by the Business Analytics Team. The engagement was undertaken in response to escalating concerns from the Operations and Finance departments regarding delivery performance variability, rising operational costs, underutilized fleet assets, and emerging safety risks across the company's nationwide freight network.

Leveraging data extracted from the company's Transportation Management System (TMS), the analytics team conducted a rigorous, multi-dimensional investigation across eight performance domains. Advanced SQL analytical techniques — including Window Functions, Common Table Expressions (CTEs), LAG-based time-series analysis, and complex multi-table joins — were applied to derive actionable business intelligence from raw transactional data.

The project delivers concrete, data-driven findings across all eight modules, summarised below:

- Driver Performance: On-time delivery rates, fuel efficiency benchmarks, and revenue-per-mile rankings were computed for all active drivers. Driver DRV00001 recorded the highest fuel efficiency of 7.00 MPG in December 2024.
- Route Profitability: Route RTE00016 emerged as the highest-revenue lane ($10.07M), while Route RTE00029 recorded the highest net profit of $21.27M. An upward fuel cost trend was identified on Route RTE00001.
- Fleet Utilization: Truck TRK00055 drove the highest total mileage (1,417,530 miles), while Truck TRK00044 generated the highest revenue ($3.05M). Truck TRK00040 was identified as operating below the fleet average.
- Maintenance Impact: Truck TRK00003 recorded the highest maintenance cost ($90,161.42) and the highest downtime (1,133.1 hours). Truck TRK00062 showed the highest recent downtime at 329.9 hours over six months.
- Fuel Efficiency: Route RTE00013 achieved the highest average MPG of 6.54. Route RTE00015 recorded the highest fuel cost per mile at $5.20. A notable fueling concentration at 12:00 AM was identified.
- Customer Revenue: XYZ Foods led revenue contribution at $1,544,419.81, while First Group recorded the highest average revenue per load at $3,294.30.
- Safety Risk: Driver DRV00058 recorded the highest incident count (7) and rate (1.06 per 100 trips). DOT Violations were the most frequent incident type with 39 occurrences. Incident volume increased slightly in 2024.
- Seasonal Demand: Shipment volumes remained stable across 2022–2024, with Spring/Summer contributing the largest seasonal revenue share (33.66%). Peak months shifted year-over-year, requiring dynamic planning.

All analytical work was completed ahead of the project deadline of March 10, 2026. The findings contained herein are intended to inform strategic decisions across Operations, Finance, Fleet Management, Sales, Compliance, and Executive leadership.

2. Project Overview

The Business Analytics Team was assigned this project as part of an internal operational intelligence initiative. The company, a third-party logistics provider managing nationwide freight operations, required a structured analytical assessment of its Transportation Management System (TMS) data to support improved decision-making across multiple business functions.

The project was designed to mirror real-world data analyst responsibilities within a 3PL environment, encompassing data extraction, multi-table join design, metric derivation, time-series analysis, and the formulation of business insights from raw operational records.

2.1. Scope & Objectives

The project scope encompassed eight analytical modules, each commissioned by a distinct stakeholder group and mapped to a specific operational concern. The table below summarises module ownership and focus areas:

Module	Stakeholder Group	Core Analytical Question
Driver Performance	Operations Leadership	Which drivers are performing at or above fleet benchmarks?
Route Profitability	Finance Department	Which freight lanes are generating profit vs. cost overruns?
Fleet Utilization	Fleet Management	Are company-owned trucks being efficiently deployed?
Maintenance Impact	Maintenance & Operations	What is the financial and operational cost of truck downtime?
Fuel Efficiency	Transportation Planning	Which routes exhibit inefficient fuel consumption?
Customer Revenue	Sales Leadership	Which customers contribute the greatest revenue value?
Safety Risk	Compliance & Safety	Where are preventable incident rates highest?
Seasonal Demand	Strategic Planning	When does demand peak, and how does it shift year-over-year?

4. Data Architecture & Sources

All analytical work was performed against the company's TMS relational database. The core schema consisted of seven primary tables spanning transactional, master, and operational data domains. The entity-relationship structure and join paths used throughout the project are detailed below:

Table	Domain	Primary Key	Key Fields	Joined To
trips	Transportation	trip_id	truck_id, driver_id, dispatch_date, miles	delivery_events, fuel_purchases, loads, maintenance_records, safety_incidents
loads	Freight	load_id	customer_id, load_date, revenue, route_id	trips, customers
delivery_events	Delivery	event_id	trip_id, on_time_flag, delivery_date	trips
fuel_purchases	Fuel Mgmt.	purchase_id	trip_id, gallons, cost, purchase_time	trips
maintenance_records	Maintenance	record_id	truck_id, cost, downtime_hrs, maint_date	trips (via truck_id)
safety_incidents	Safety	incident_id	trip_id, preventable_flag, incident_type, severity	trips
customers	CRM	customer_id	customer_name, region, segment	loads

All join paths were designed to maintain referential integrity across transactional and master data layers, ensuring accurate aggregation at driver, route, truck, customer, and time-period levels.

5. Analytical Framework

The project applied a four-pillar SQL analytics framework, selected to address the varying complexity of each business question. The techniques are described below alongside their application context:

Technique	Description	Modules Applied
Window Functions	Applied RANK(), DENSE_RANK(), NTILE(), PERCENT_RANK(), FIRST_VALUE(), SUM() OVER, and AVG() OVER to rank, compare, and partition metrics within peer groups without collapsing result sets.	All 8 Modules
Common Table Expressions (CTEs)	Used to decompose multi-step analytical logic into modular, readable sub-queries — enabling fleet average comparisons, share calculations, and filtered aggregations.	Modules 1, 2, 3, 5, 6, 7, 8
LAG / Time-Series Analysis	Employed the LAG() function alongside monthly date partitioning to detect Month-over-Month trends, growth rates, and directional cost movements.	Modules 1, 2, 3, 5, 6, 8
Multi-Table Joins	Constructed INNER and LEFT JOINs across 2–4 tables per query to derive cross-domain metrics such as maintenance cost vs. revenue and incident rate per 100 trips.	All 8 Modules

MOD 01 DRIVER PERFORMANCE INSIGHTS

Business Requirement

Operations leadership required a comprehensive evaluation of driver-level performance to identify high-performing drivers and isolate those associated with delivery delays or operational inefficiencies. Metrics were required at individual driver level as well as relative to fleet benchmarks.

Key Metrics Derived

- On-Time Delivery Rate per Driver
- Average Fuel Efficiency (MPG) per Driver
- Revenue Generated per Mile Driven per Driver
- Month-over-Month Revenue Trend per Driver

Analytical Techniques

- RANK() OVER (PARTITION BY month ORDER BY on_time_rate DESC) — Monthly on-time performance ranking
- NTILE(10) / PERCENT_RANK() — Identification of top 10% drivers by revenue per mile
- CTE + AVG() OVER () — Driver MPG vs. fleet average comparison
- LAG() OVER (PARTITION BY driver_id ORDER BY month) — Month-over-Month revenue trend

Operational Findings

On-Time Delivery Performance

An analysis of delivery event records against scheduled delivery windows was conducted across all active drivers. The fleet-wide on-time delivery rate was established as the primary benchmark for driver reliability assessment.

55.67%	58.98%
Fleet Average On-Time Rate	DRV00108 On-Time Rate
Fleet Benchmark	*+3.31% above average*

Driver **DRV00108** achieved an **on-time delivery rate** of **58.98%,** positioning this driver above the **fleet average of 55.67%** and indicating comparatively stronger delivery reliability. Monthly ranking analysis (Window Function) enables operations managers to monitor rank movement across reporting periods and intervene where performance deteriorates.

Revenue per Mile Performance

Revenue generation efficiency was assessed by calculating the revenue generated per mile driven at the individual driver level, providing a composite indicator of route profitability and operational effectiveness.

$2.20
Highest Rev/Mile — DRV00101
per mile driven

Driver **DRV00101** recorded the highest revenue per mile at **$2.20**, indicating strong route profitability and operational efficiency relative to peers. The top 10% of drivers by this metric were identified using NTILE(10) partitioning, providing a high-performance cohort for preferential load assignment consideration.

Fuel Efficiency (MPG) Analysis

Fuel efficiency was evaluated at the individual driver level and benchmarked against the fleet average MPG. A CTE was used to compute the fleet average, which was then compared against each driver's observed MPG via a Window Average function.

Metric	Value	Observation
Fleet Average MPG	6.50 MPG	Operational benchmark for all driver comparisons
Driver DRV00017 — Average MPG	6.52 MPG	+0.02 MPG above fleet average; marginally more fuel-efficient
Highest MPG — December 2024	7.00 MPG (Driver DRV00001)	Best fuel performance in the most recent reporting period

Driver **DRV00017 recorded an average fuel efficiency of 6.52 MPG**, marginally **exceeding the fleet average of 6.50 MPG**. In the most recent observed month of December 2024, Driver DRV00001 achieved the highest fuel efficiency at 7.00 MPG, representing the strongest single-driver fuel performance in the dataset for the latest reporting period. This finding is significant for benchmarking and identifying fuel-optimal driving behaviours that may be replicated across the fleet.

> **Brain Teaser — Resolved:** *Which driver had the highest MPG in the most recent month? Driver DRV00001 recorded the highest fuel efficiency at 7.00 MPG in December 2024, identified via RANK() OVER (PARTITION BY month ORDER BY avg_mpg DESC) filtered to the most recent dispatch period.*

Data Sources

Tables: trips, delivery_events, loads

Join Path: trips → delivery_events (trip_id) | trips → loads (load_id)

MOD 02 ROUTE PROFITABILITY ASSESSMENT

Business Requirement

Finance stakeholders required visibility into which freight lanes are generating profit and which are incurring cost overruns driven by fuel expenditure. The analysis was required to support pricing reviews, route rationalisation decisions, and operational cost management.

Key Metrics Derived

- Total Revenue per Route
- Total Fuel Cost per Route
- Net Route Profitability (Revenue minus Fuel Cost)
- Monthly Profit Trend per Route

Analytical Techniques

- Time Series: Monthly profit trend using date-based GROUP BY aggregation
- DENSE_RANK() OVER (PARTITION BY quarter ORDER BY net_profit DESC) — Top 5 routes per quarter
- LAG() OVER (PARTITION BY route_id ORDER BY month) — Three-month fuel cost escalation detection
- CTE: Route profitability vs. company average benchmark comparison

Operational Findings

Revenue Performance

Total revenue was aggregated at the route level by joining load revenue records to their corresponding trip and fuel purchase records. This enabled a direct comparison of revenue generation and fuel cost across all active transportation lanes.

$10.07M	$353,891.77	$8.77M
Highest Revenue Route	Highest Dec 2024 Revenue	Company Avg Route Profit
Route RTE00016	*Route RTE00016*	*Benchmark*

Route **RTE00016** generated the highest **total revenue of $10,069,054.22**, establishing it as the most commercially significant transportation lane in the network. The **same route** also recorded the highest monthly revenue in **December 2024 at $353,891.77**, confirming sustained high performance on this lane.

Fuel Cost Analysis

Fuel costs were computed at the route level by joining trip fuel purchase records to their parent loads via trip_id. This enabled identification of routes where fuel expenditure represents a disproportionate share of revenue.

$1.79M
Highest Fuel Cost Route
Route RTE00047

Route **RTE00047** incurred the highest **total fuel cost at $1,786,126.58**, indicating **above-average** operating expenses relative to other lanes. This route warrants review of load density, distance optimisation, and fuel rate agreements.

Net Route Profitability

Net profitability was calculated as the difference between total route revenue and total route fuel costs. Results were benchmarked against the company-wide average route profit using a CTE.

Route	Net Profit	vs. Company Avg ($8,767,805.44)	Assessment
RTE00016	$8,460,671.82	-$307,133.62	Slightly below average; high revenue, manageable costs
RTE00029	$21,269,491.08	+$12,501,685.64	Significantly above average; highest net profit lane
Company Average	$8,767,805.44	— Benchmark —	Reference figure for route comparison

Route **RTE00029** recorded an exceptionally high **net profit of $21,269,491.08**, substantially **exceeding the company average** route profit of **$8,767,805.44**. This route should be prioritised for capacity allocation and protected from pricing erosion. Route RTE00016, while generating the highest absolute revenue, yielded a net profit marginally below the company average, indicating scope for fuel cost optimisation.

Fuel Cost Trend — Route RTE00001

A three-month LAG analysis was applied to identify routes exhibiting a consistent upward fuel cost trajectory. Route RTE00001 was identified as showing a sustained escalation in monthly fuel expenditure:

Period	Monthly Fuel Cost ($)	Month-over-Month Change
Month 1 (Base)	$48,544.78	—
Month 2	$53,390.79	+$4,846.01 (+9.98%)
Month 3 (Most Recent)	$56,680.25	+$3,289.46 (+6.16%)

The steady three-month increase in fuel costs on Route RTE00001 signals rising operational cost pressures on this lane and warrants immediate investigation into load optimisation, driver fuel behaviour, or renegotiation of fuel surcharge agreements with customers.

Brain Teaser — Resolved: *Which route generated the highest revenue last month? Route RTE00016 recorded the highest monthly revenue of $353,891.77 in December 2024, identified by filtering load records to the most recent month and ranking routes by SUM(revenue) DESC.*

Data Sources

Tables: loads, trips, fuel_purchases

Join Path: loads → trips (load_id) | trips → fuel_purchases (trip_id)

MOD 03 FLEET UTILIZATION MONITORING

Business Requirement

Fleet management required an assessment of whether company-owned trucks are being deployed efficiently and contributing proportionally to revenue generation. The analysis was designed to support asset reallocation, retirement planning, and utilisation benchmarking.

Key Metrics Derived

- Total Miles Driven per Truck
- Total Revenue Generated per Truck
- Monthly Revenue per Truck
- Rolling 3-Month Utilisation per Truck

Analytical Techniques

- SUM() OVER (PARTITION BY truck_id ROWS BETWEEN 2 PRECEDING AND CURRENT ROW) — Rolling 3-month mileage
- RANK() OVER (PARTITION BY month ORDER BY revenue DESC) — Revenue rank within each month
- CTE: Identification of trucks operating below fleet average mileage
- LAG() OVER (PARTITION BY truck_id ORDER BY month) — Monthly utilisation growth rate

Operational Findings

Distance Utilisation

Total miles driven was computed per truck across all recorded trips. Fleet average mileage was computed via CTE and used as the benchmark for under-utilisation identification.

1,417,530	1,313,540	1,178,515
Highest Mileage — TRK00055	Fleet Average Mileage	TRK00040 Mileage
total miles driven	*miles per truck*	*below fleet average*

Truck **TRK00055** recorded the highest **total distance driven at 1,417,530 miles**, indicating heavy operational deployment. In contrast, **Truck TRK00040 recorded 1,178,515 miles driven** — 134,024 miles below the **fleet average of 1,313,539.80 miles** — identifying it as an underutilised asset. Fleet management should evaluate whether TRK00040's lower mileage reflects route assignment patterns, mechanical reliability concerns, or scheduling inefficiencies.

Revenue Generation by Truck

Revenue was attributed to each truck by joining trip records to their associated loads. This metric reflects both the volume and commercial value of loads assigned to each asset.

$3,045,707.57
Highest Revenue Truck
Truck TRK00044

Truck **TRK00044** generated the **highest total revenue of $3,045,707.57**, establishing it as the most revenue-productive asset in the fleet. Monthly revenue ranking analysis (RANK() OVER PARTITION BY month) enables fleet management to monitor the revenue contribution rank of each truck within each reporting period and respond to sustained underperformance.

Fleet Activity Growth

Month-over-Month mileage changes were analysed using the LAG() function to identify trucks with expanding operational deployment.

Truck	Previous Period Miles	Miles Growth	Assessment
TRK00070	60,809 miles	+38,674 miles	Significant expansion in operational deployment — monitor for asset wear

Truck TRK00070 exhibited a growth of 38,674 miles in operational distance from its previously recorded 60,809 miles. This significant increase warrants monitoring to ensure maintenance schedules are aligned with the elevated utilisation pace.

Data Sources

Tables: trips, loads

Join Path: trips → loads (load_id)

MOD 04 MAINTENANCE COST IMPACT ANALYSIS

Business Requirement

The Maintenance and Operations teams required a quantitative assessment of the financial and operational impact of truck maintenance activities. The analysis aimed to identify high-cost, high-downtime assets and evaluate how maintenance events affect trip volume and revenue generation.

Key Metrics Derived

- Total Maintenance Cost per Truck
- Total Downtime Hours per Truck
- Maintenance Cost per Mile Driven
- Downtime Impact on Monthly Trip Count

Analytical Techniques

- Time Series: Maintenance cost trend over time using aggregation
- RANK() OVER (PARTITION BY last_6_months ORDER BY downtime DESC) — Top 5 high-downtime trucks
- CTE + JOIN: Downtime hours joined to monthly trip counts to quantify operational impact
- JOIN: Side-by-side maintenance cost vs. revenue comparison per truck

Operational Findings

Highest Maintenance Cost & Downtime

Maintenance expenditure and associated downtime hours were aggregated at the truck level from maintenance_records. The truck-level comparison revealed a concentration of both cost and downtime in a single asset.

$90,161.42	1,133.1 hrs	$0.0579/mi
Highest Maintenance Cost	Highest Total Downtime	Highest Cost/Mile
Truck TRK00003	*Truck TRK00003*	*Truck TRK00014*

Truck **TRK00003** incurred the **highest maintenance expenditure at $90,161.42** and simultaneously recorded the **highest total downtime of 1,133.1 hours**. The co-occurrence of maximum maintenance cost and maximum downtime on a single asset indicates a potential correlation between chronic mechanical issues and asset availability constraints. This truck should be assessed for comprehensive overhaul or fleet replacement prioritisation.

Separately, **Truck TRK00014** recorded the **highest maintenance cost per mile at $0.0579 per mile**, suggesting elevated maintenance intensity relative to its operational output. High cost-per-mile assets effectively compress the margin on every load they haul and should be targeted in fleet renewal planning.

Recent Downtime — Last Six Months

A rolling six-month downtime analysis (ending 31 December 2024) was conducted using a window function to identify trucks with the highest recent downtime exposure.

329.9 hrs
Highest Recent Downtime (6M)
Truck TRK00062

Within the six-month window ending **31 December 2024**, Truck **TRK00062** recorded **the highest downtime at 329.9 hours**, indicating significant recent operational disruptions. This truck should be placed on a priority maintenance review schedule ahead of the next peak demand period.

Downtime vs. Trip Count Comparison

A CTE-based join was constructed to compare total trip count against total downtime hours, quantifying the operational opportunity cost of maintenance events.

Truck	Total Trips Completed	Total Downtime (Hours)	Observation
TRK00055	977 trips	689.5 hours	Highest trip count; significant downtime despite high utilisation
TRK00079	831 trips	572.4 hours	Lower trip count; proportionally lower downtime

Truck **TRK00055** completed the **highest number of trips at 977**, yet also experienced substantial **downtime of 689.5 hours** — suggesting that high utilisation is occurring alongside maintenance events rather than being prevented by them. Truck **TRK00079** completed fewer trips with comparatively lower downtime, indicating a different utilisation profile. These comparisons support the need for predictive maintenance scheduling to minimise unplanned downtime on high-utilisation assets.

> **Brain Teaser — Resolved:** *Which truck had zero downtime last quarter? Identified via a LEFT JOIN between trucks and maintenance_records filtered to Q4 2024, returning trucks with NULL or zero recorded downtime values. Trucks not present in maintenance_records for the period were confirmed as having zero downtime.*

Data Sources

Tables: maintenance_records, trips

Join Path: maintenance_records → trips (truck_id)

MOD 05 FUEL EFFICIENCY ANALYSIS

Business Requirement

The transportation planning team required an analysis of fuel consumption patterns at the route level to identify inefficient lanes and inform re-routing, driver training, and fuel cost management strategies.

Key Metrics Derived

- Average MPG per Route
- Fuel Cost per Mile per Route
- MPG Trend per Route by Month
- Rolling Average MPG per Route

Analytical Techniques

- Time Series: Monthly MPG trend per route using dispatch_date aggregation
- RANK() OVER (ORDER BY avg_mpg DESC) — Route-level fuel efficiency ranking
- CTE: Identification of routes with MPG below fleet median
- AVG() OVER (PARTITION BY route_id ROWS BETWEEN 2 PRECEDING AND CURRENT ROW) — Rolling average MPG

Operational Findings

Route-Level Fuel Efficiency

Average miles per gallon was computed at the route level by joining trip distance records to fuel purchase quantities. Routes were ranked by fuel efficiency to identify best-in-class and underperforming lanes.

6.54 MPG	$5.20/mile
Most Fuel-Efficient Route	Highest Fuel Cost/Mile
Route RTE00013	*Route RTE00015*

Route **RTE00013** recorded the **highest average fuel efficiency at 6.54 MPG**. Routes RTE00013 and RTE00045 jointly represent the most fuel-efficient transportation lanes in the network and provide a benchmark for efficiency-focused routing decisions.

In contrast, Route **RTE00015 recorded the highest fuel cost per mile at $5.20 per mile**, indicating potential inefficiencies or particularly demanding operating conditions on this lane. This route warrants a detailed review of vehicle assignments, load density, and route geometry.

Fleet Median Fuel Benchmark

Routes were assessed relative to the fleet median fuel metric to identify lanes operating at, above, or below the network-wide central tendency.

Route	Fuel Metric	vs. Fleet Median (9,560.98)	Assessment
RTE00009	9,559.48	-1.50	Marginally below fleet median
RTE00013	9,558.17	-2.81	Marginally below fleet median
RTE00016	9,555.62	-5.36	Close to fleet median; broadly efficient
Fleet Median	9,560.98	— Benchmark —	Reference metric for route efficiency comparison

Routes **RTE00009, RTE00013, and RTE00016** all perform in **close proximity to the fleet median fuel benchmark of 9,560.9**8. This indicates that these routes exhibit broadly consistent and efficient fuel consumption relative to the network average.

Fuel Purchase Timing Patterns

An analysis of fuel purchase timestamps revealed a notable concentration of refuelling activity at specific hours of the day.

12:00 AM
Peak Fueling Hour
10,848 transactions recorded

The **highest volume of fuel purchases — totalling 10,848 recorded transactions — occurred at 12:00 AM**. This concentration of overnight fuelling activity may reflect structured depot refuelling protocols, but warrants investigation to ensure fuel card controls and authorisation procedures are enforced during low-supervision overnight hours.

Data Sources

Tables: trips, loads, fuel_purchases

Join Path: loads → trips (load_id) | trips → fuel_purchases (trip_id)

MOD 06 CUSTOMER REVENUE ANALYSIS

Business Requirement

Sales leadership required identification of the company's highest-value customers and an assessment of customer-level revenue contribution relative to the total company portfolio. The analysis was designed to support account management prioritisation, upsell targeting, and concentration risk evaluation.

Key Metrics Derived

- Total Revenue per Customer
- Average Revenue per Load per Customer
- Monthly Revenue Trend per Customer
- Customer Revenue Share vs. Total Company Revenue

Analytical Techniques

- Time Series: Monthly revenue trend per customer using load_date aggregation
- DENSE_RANK() OVER (PARTITION BY quarter ORDER BY revenue DESC) — Top customers by quarter
- CTE: Customer revenue as a percentage of total — SUM() OVER () for grand total denominator
- LAG() OVER (PARTITION BY customer_id ORDER BY month) — Month-over-Month revenue growth rate

Operational Findings

Top Revenue-Contributing Customers

Total revenue was aggregated at the customer level by joining load revenue records to customer master data. Year-to-date rankings were computed to identify the company's highest-value accounts.

Rank	Customer	Total Revenue ($)	Revenue Share (%)	Assessment
1	XYZ Foods	$1,544,419.81	0.59% of total	Highest revenue contributor — top-priority account
2	Superior Group	$1,542,321.02	0.59% of total	Second highest — near-equal contribution to XYZ Foods
—	First Logistics	$309,543.23	Lowest share	Lowest revenue contributor — lowest-value account in dataset

XYZ Foods emerged as the largest revenue contributor at $1,544,419.81, closely **followed by Superior Group at $1,542,321.02**. Both customers account for approximately **0.59% of total company revenue each**, representing the highest individual customer revenue shares in the dataset. The narrow gap between these two accounts suggests a competitive and well-balanced top-tier customer portfolio.

First Logistics generated the **lowest total revenue contribution at $309,543.23**. This account should be reviewed to determine whether it represents a growth opportunity, a low-margin legacy relationship, or a candidate for account rationalisation.

Revenue per Load Performance

Average revenue per load was computed to assess the shipment value efficiency of each customer account, independent of total volume.

$3,294.30
Highest Rev/Load
First Group (avg per load)

First Group recorded the highest average revenue per load at $3,294.30, indicating strong shipment value per transaction. Despite not appearing in the top total revenue rankings, this account's high per-load value positions it as a premium-yield customer warranting strategic attention from the sales team.

> **Brain Teaser — Resolved:** *Which customer contributed the most revenue this year? XYZ Foods recorded the highest year-to-date revenue of $1,544,419.81, identified by filtering loads to the current year and ranking customers by SUM(revenue) DESC.*

Data Sources

Tables: customers, loads

Join Path: customers → loads (customer_id)

MOD 07 SAFETY RISK ASSESSMENT

Business Requirement

The Compliance and Safety departments required a data-driven assessment of incident occurrence patterns across the driver population, with particular focus on preventable accident rates, incident type distribution, and temporal risk patterns to support targeted intervention and regulatory compliance.

Key Metrics Derived

- Incident Count per Driver
- Preventable Incident Rate per Driver
- Monthly and Annual Incident Trend
- Incident Rate per 100 Trips

Analytical Techniques

- Time Series: Monthly incident trend using incident_date aggregation
- RANK() OVER (ORDER BY incident_count DESC) — Driver ranking by incident frequency
- CTE: Preventable vs. non-preventable incident ratio computation
- JOIN: safety_incidents to trips for incident rate per 100 trips calculation

Operational Findings

Driver Incident Analysis

Incident records were joined to trip data to compute driver-level incident counts and frequency rates. Drivers were ranked by incident frequency and preventable incident rate to identify high-risk individuals.

7 incidents	1.06 per 100 trips	DRV00138
Highest Incident Count	Highest Incident Rate	100% Preventable Rate
Driver DRV00058	*Driver DRV00058 (660 trips)*	*All incidents preventable*

Driver **DRV00058** recorded the **highest number of incidents at 7** and the **highest incident rate of 1.06** incidents per 100 trips, based on **660 total trips completed**. This driver represents the highest safety risk profile in the fleet and should be prioritised for immediate compliance review and targeted retraining.

Driver DRV00138 recorded a 100% preventable incident rate, indicating that all incidents associated with this driver were classified as preventable under company safety protocols. Preventable incidents directly expose the organisation to regulatory liability and insurance cost increases, making this driver a priority candidate for behavioural safety intervention.

Annual Incident Trend

Incident records were aggregated annually to identify long-term trends in fleet safety performance.

Year	Total Incidents	Year-over-Year Change	Assessment
2022	56	—	Baseline year
2023	54	-2 (-3.6%)	Marginal improvement
2024	60	+6 (+11.1%)	Increase — warrants investigation

The annual incident data reveals a **slight reduction from 56 incidents in 2022 to 54 in 2023**, followed by an **increase to 60 incidents in 2024** — representing an **11.1% rise r**elative to the prior year. The reversal of the improving trend in 2024 requires root cause analysis to determine whether it reflects fleet expansion, route complexity changes, or deteriorating safety culture.

Preventable vs. Non-Preventable Breakdown

All incidents were classified as preventable or non-preventable. A CTE was used to compute the ratio and percentage contribution of each category.

Category	Incident Count	Share of Total	Implication
Preventable	64	37.6%	Attributable to controllable operational factors
Non-Preventable	106	62.4%	External or unavoidable events
Total	170	100%	Preventable ratio: 0.60 (preventable per non-preventable)

The preventable incident ratio of 0.60 indicates that for every non-preventable incident, approximately 0.60 preventable incidents occur. With 64 preventable incidents on record, there is a meaningful opportunity to reduce total incident volume through targeted driver behaviour programmes, as preventable events are by definition within the organisation's control.

Accident Timing Patterns

Incident timestamps were analysed to identify periods of elevated accident risk within the operational day.

Peak Incident Hour	Incident Count	Risk Implication
17:00 (5:00 PM)	12 incidents	Evening shift changeover / peak-hour traffic exposure
03:00 (3:00 AM)	12 incidents	Overnight fatigue risk — low-visibility driving conditions

Two distinct **peak accident periods** were identified at **17:00 and 03:00 hours**, each recording 12 incidents. The 17:00 concentration may reflect evening shift transition pressure and peak traffic conditions, while the 03:00 peak is consistent with driver fatigue risk during overnight operations. These findings support the introduction of structured fatigue management protocols and enhanced monitoring during these windows.

Most Common Incident Types

Incidents were categorised by type to identify the predominant safety risk areas requiring intervention.

Incident Type	Occurrences	Severity Implication
DOT Violations	39	Regulatory compliance risk — highest frequency
Accidents	35	Direct safety and liability exposure
Equipment Damage	35	Asset cost impact and potential safety risk

DOT Violations represent the most frequent incident type at 39 occurrences, posing the greatest regulatory compliance risk. The equal frequency of Accidents and Equipment Damage incidents (35 each) indicates a need for parallel programmes addressing both driver safety behaviour and vehicle pre-trip inspection rigour.

Incident Severity Distribution

Severity Level	Total Incidents	% of Total	Action Priority
Moderate	62	36.5%	Largest category — process review recommended
Severe	55	32.4%	High impact — immediate intervention warranted
Minor	53	31.2%	Lowest category — monitor for escalation patterns
Total	170	100%	—

Moderate incidents constitute the largest share of reported safety events at 62 occurrences (36.5%), followed closely by Severe incidents at 55 (32.4%). The near-equal distribution across severity levels underscores the need for a comprehensive safety programme rather than one focused exclusively on the most critical events.

Data Sources

Tables: safety_incidents, trips

Join Path: safety_incidents → trips (trip_id)

MOD 08 SEASONAL DEMAND PATTERN ANALYSIS

Business Requirement

The strategic planning team required identification of seasonality trends in shipment volume and revenue to support demand forecasting, driver scheduling, and capacity allocation decisions across the fiscal year.

Key Metrics Derived

- Monthly and Annual Load Volume
- Monthly Revenue Trend
- Monthly Trip Volume
- Quarter-wise Shipment Trend

Analytical Techniques

- LAG() OVER (ORDER BY month) — Month-over-Month load volume growth rate
- FIRST_VALUE() / RANK() OVER (PARTITION BY year ORDER BY load_count DESC) — Peak month identification
- CTE: Revenue contribution bucketed by season (Spring/Summer, Monsoon/Rain, Autumn/Winter)
- Time Series: Quarter-wise aggregation using load_date and dispatch_date

Operational Findings

Annual Trip Volume

Total trip volume was aggregated annually across the three-year observation period to assess the overall stability of operational demand.

Year	Total Trips	Year-over-Year Change	Assessment
2022	28,589	—	Baseline year
2023	28,165	-424 (-1.5%)	Marginal decline
2024	28,656	+491 (+1.7%)	Recovery to near-2022 levels

Shipment volume remained relatively **stable across the three-year** observation period, ranging from **28,165 to 28,656 trips annually**. The marginal decline in 2023 was largely recovered in 2024, suggesting a resilient demand base with limited year-over-year variability. This stability enables reliable capacity planning based on historical averages.

Peak Shipment Months

Peak months were identified annually using a window function to rank months by total load volume within each year.

Year	Peak Month	Total Shipments	Observation
2022	August	2,496	Summer demand peak
2023	October	2,474	Autumn demand peak — seasonal shift
2024	January	2,437	Winter demand peak — further seasonal shift

The analysis reveals a consistent shift in peak demand months across consecutive years — from August 2022, to October 2023, to January 2024. This year-over-year migration of the peak period indicates that the company cannot rely on a fixed seasonal planning assumption and must apply dynamic forecasting to accurately anticipate peak capacity requirements. Strategic planning should incorporate rolling 12-month trend analysis to stay ahead of these shifts.

Revenue Contribution by Season

Revenue was segmented into three operational seasons using a CTE-based bucketing approach to evaluate the commercial weight of each seasonal period.

Season	Seasonal Revenue ($)	Revenue Share (%)	Assessment
Spring / Summer	$88,357,917.51	33.66%	Highest seasonal revenue contributor
Monsoon / Rain	$88,206,537.40	33.60%	Closely aligned with Spring/Summer
Autumn / Winter	$85,961,345.38	32.74%	Marginally lowest — but broadly balanced
Total	$262,525,800.29	100%	—

Seasonal revenue distribution is broadly balanced across all three operational periods, with **Spring/Summer contributing the largest share at 33.66%** ($88,357,917.51), followed by the **Monsoon/Rain season at 33.60%** ($88,206,537.40) and **Autumn/Winter at 32.74%** ($85,961,345.38).

> **Brain Teaser — Resolved:** *Which month had the highest load volume? August 2022 recorded the highest single-month load volume of 2,496 shipments across the full dataset, identified via RANK() OVER (PARTITION BY year ORDER BY load_count DESC) with the result filtered to rank = 1.*

Data Sources

Tables: loads, trips

Join Path: Time-based aggregation using load_date and dispatch_date

14. Cross-Module Key Findings

The following synthesised findings represent the most significant cross-module insights derived from the eight analytical workstreams. These findings are intended to inform executive-level decision-making and cross-functional action planning.

Driver & Delivery Performance

- Fleet on-time delivery rate of 55.67% indicates that nearly half of all deliveries are not meeting scheduled windows — a systemic performance challenge requiring structured intervention.
- Driver DRV00101's revenue-per-mile of $2.20 significantly exceeds peer performance, validating route-optimised load assignment as a revenue lever.
- The top 10% of drivers by revenue per mile generate disproportionately higher returns; preferential load assignment to this cohort is recommended.

Route Economics & Fuel Management

- Route RTE00029 generates $21.27M in net profit — 2.43x the company average — and should be protected through dedicated capacity allocation and contract security.
- The three-month upward fuel cost trajectory on Route RTE00001 (+16.8% cumulative increase) requires immediate operational review to prevent margin erosion.
- Route RTE00015's fuel cost per mile of $5.20 is an outlier requiring root-cause investigation into vehicle assignment, load density, or route geometry.

Fleet Asset Management

- Truck TRK00003 represents a concentrated maintenance risk: highest total maintenance cost ($90,161.42) and highest total downtime (1,133.1 hours) on a single asset.
- TRK00040 operating 134,024 miles below fleet average indicates meaningful underutilisation that should be remediated through route reassignment.
- TRK00070's rapid mileage growth (+38,674 miles) warrants proactive maintenance scheduling to prevent the asset from mirroring the high-downtime profile of older trucks.

Customer Portfolio

- XYZ Foods and Superior Group collectively represent the highest revenue accounts yet each account for only 0.59% of total revenue — signalling a highly fragmented customer base with limited concentration risk, but also limited key-account leverage.
- First Group's high average revenue per load ($3,294.30) despite lower absolute revenue positions it as a strategically valuable account for volume growth targeting.

Safety & Compliance

- The 11.1% increase in total incidents in 2024 (60 vs. 54 in 2023) is the most critical fleet-wide safety signal and demands an immediate programme review.
- DOT Violations are the most frequent incident type (39 occurrences), presenting the highest regulatory risk exposure in the fleet.

- The dual 12:00 AM accident peaks (midnight fueling and 03:00 hours fatigue incidents) suggest overnight operations are disproportionately high-risk and require enhanced protocols.

Demand Planning

- The shifting peak month — from August to October to January across 2022–2024 — confirms that static seasonal planning models are no longer adequate for this operation.
- The near-uniform seasonal revenue split (33.66% / 33.60% / 32.74%) suggests limited seasonal pricing power; dynamic rate structures should be explored to capture yield during demand peaks.

15. Strategic Recommendations

The following recommendations are derived directly from the analytical findings across all eight modules. Each recommendation is assigned a priority level (High / Medium / Low), a responsible owner, and an expected business impact.

Priority	Recommendation	Owner	Module(s)	Expected Impact
HIGH	Initiate immediate safety programme review targeting 2024's 11.1% incident increase; implement fatigue management protocols for 03:00 and 17:00 operational windows.	Compliance & Safety	Module 7	Reduce incident rate; lower insurance and regulatory exposure
HIGH	Place Driver DRV00058 and Driver DRV00138 on mandatory safety retraining; apply enhanced monitoring for all drivers in the top quartile of preventable incident rate.	Operations / HR	Module 7	Reduce preventable incident rate toward zero
HIGH	Investigate and address upward fuel cost trend on Route RTE00001 — review load assignments, driver behaviour, and customer rate agreements.	Finance / Ops	Module 2	Arrest fuel cost escalation; protect route margin
HIGH	Assess Truck TRK00003 for comprehensive overhaul or fleet replacement; its combined maintenance cost and downtime profile indicates near end-of-useful-life status.	Fleet Management	Module 4	Reduce unplanned downtime; lower cost per trip
MEDIUM	Prioritise Truck TRK00062 for pre-season maintenance review given 329.9 hours of recent downtime; schedule before the next identified demand peak.	Fleet Management	Module 4	Avoid peak-season service disruptions
MEDIUM	Reassign Truck TRK00040 to higher-demand routes to close its 134,024-mile gap versus fleet average; review scheduling logic.	Fleet Management	Module 3	Improve fleet-wide utilisation rate
MEDIUM	Develop key-account growth plans for XYZ Foods and Superior Group to deepen relationship value beyond current 0.59% revenue share each.	Sales	Module 6	Reduce customer concentration risk; grow top-line
MEDIUM	Adopt dynamic peak-month forecasting using rolling 12-month LAG analysis; discontinue static seasonal planning assumptions given year-over-year peak migration.	Strategic Planning	Module 8	Improve capacity accuracy; reduce over/under-staffing

Priority	Recommendation	Owner	Module(s)	Expected Impact
LOW	Establish route-level MPG deviation alerts (>15% below fleet benchmark); investigate Route RTE00015's $5.20/mile fuel cost.	Transportation Planning	Module 5	Reduce fuel spend through targeted optimisation
LOW	Review overnight fuel authorisation controls given 10,848 fuel transactions recorded at 12:00 AM; implement time-based fuel card restrictions.	Finance / Security	Module 5	Reduce fraud risk; improve fuel governance

16. Conclusion

This report presents the complete findings and strategic recommendations of the Logistics Operations Analytics project. Across eight analytical modules, the Business Analytics Team has successfully transformed raw Transportation Management System data into a structured, multi-dimensional intelligence asset that equips the organisation's leadership with the evidence base required to make data-driven operational and strategic decisions.

The analyses conducted — spanning driver performance, route profitability, fleet utilisation, maintenance impact, fuel efficiency, customer revenue, safety risk, and seasonal demand — collectively reveal an organisation with strong performance in specific lanes and assets, alongside identifiable and addressable gaps in safety compliance, fleet asset health, and fuel cost management.

The most urgent findings centre on the 2024 safety incident increase, the concentrated maintenance and downtime profile of Truck TRK00003, and the escalating fuel cost trajectory on Route RTE00001. These three areas represent the highest-priority interventions and should be actioned within the immediate post-report period.

Medium-term priorities include fleet rebalancing, customer account development, and the transition to dynamic seasonal forecasting models. Long-term, the organisation should consider investment in predictive maintenance scheduling and route-level fuel optimisation analytics as embedded operational capabilities rather than periodic assessments.

It is recommended that this report be reviewed in a cross-functional leadership session and that named owners be assigned to each High and Medium priority recommendation. A 90-day post-implementation review should be scheduled to measure the impact of all interventions.

The Business Analytics Team remains available to support deeper analysis on any individual module, develop executive dashboards from these findings, or assist with implementation planning as required.

— End of Report —

Business Analytics Team | Third-Party Logistics Company | March 10, 2026